

11.0 Start Up: Electrical Verifications (Panel Disconnect in the OFF Position)



Prevent serious injury or death.

Electrical service installation must be performed by a qualified electrician and conform to all electrical codes.

The startup is performed in stages. For this first stage we are going to verify the electrical installation, 3-phase supply power, controls components, control wiring, and incoming field wiring.

1.

Using a properly rated voltmeter verify supply voltage is coming into the top side of the disconnect and that it matches the supply voltage called on the Hapman supplied electrical prints.

2.

Use voltmeter to verify supply voltage at top of the disconnect on all legs, leg to leg, and each leg to ground.

3.

Verify all circuit breakers, motor protectors, and overloads are set to the proper motor FLA per the motor(s) nameplate(s).

4.

Verify all fuses are installed and secure

5.

Verify all control relays (white part of relay) are pushed into the bases and secured.

6.

Verify all field wiring is labeled and terminated correctly per the print. *All field wiring (excluding outgoing motor wires from the VFD) should be terminated on the left side of the terminal strip and the internal wiring on the right side of the terminal strip.*

7.

Close the panel door.

NOTICE

On the hinge bar grate style Helix hopper, the “Bar Grate Limit Switch” is referred to as a “Mechanical Interlock” on the electrical prints. It will need to be wired per the prints for the unit to run. Reference appendix E in this manual for instructions.

11.0 Start Up: Electrical Verifications (Panel Disconnect in the ON Position)

 WARNING	Prevent serious injury or death. Electrical service installation must be performed by a qualified electrician and conform to all electrical codes.
1.	Turn disconnect lever to the “ON” position.
2.	Using proper PPE and safety protocols for your rated environment, open the panel door.
3.	Use a properly rated voltmeter to verify voltage at the bottom of the disconnect (leg to leg and each leg to ground)
4.	Use the voltmeter to verify that the 3 phase supply voltages called out on the electrical print are being delivered from the output side of the transformer(s).
5.	Use the voltmeter to verify that the correct voltages is coming in to and going out of all fuses.
6.	Verify that 24 VDC power supply is on. <i>(DC power supply has green “DC OK” light that will be illuminated)</i>
7.	Use the voltmeter to verify that 24vdc is being output by the DC power supply.
8.	Verify that the red safety relay has 3 green lights on (power, channel 1 and channel 2) If all three lights are not illuminated there is a component or wiring issue that must be addressed before moving on.
9.	Verify VFD does not show any active faults on the VFD display <i>(If your unit does not have a VFD you can disregard this step)</i>
10.	With panel door open, press the E-stop in on panel door.

11.0 Start Up: Electrical Verifications (Panel Disconnect in the ON Position)

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11.	Visually verify in the panel that the safety relay power light has remained green. Also, that the green channel 1 and channel 2 lights go out on the safety relay.
NOTICE	<i>If there is another safety device in series with the E-stop you will only lose a single channel instead of both. If there are no other safety devices in series with the E-stop, then you should loose both channels.</i>
12.	Once step 11 is confirmed, pull the E-stop back out and verify that the safety relay power light is still on and that both channels 1 and 2 have their green lights illuminated.
13.	If you unit includes additional E-stops or other safety sensors, then you must verify function on each one. Repeat steps 11 & 12 for each additional device until all are verified.
14.	Once all E-stops and safety devices have been verified as functional you can proceed to the next stage of startup.
NOTICE	<i>On the hinge bar grate style Helix™ hopper, the “Bar Grate Limit Switch” is referred to as a “Mechanical Interlock” on the electrical prints. It will need to be wired per the prints for the unit to run. Reference appendix E in this manual for instructions.</i>
15.	Close panel door and lock panel door.

11.0 Start Up: Conveyor Checks with No Material (Motor Rotation)



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Prevent serious injury or death.

Severing and entanglement hazard.
Keep clear of conveyor and all moving parts.

The startup is performed in stages. For the second stage we are going to verify correct motor rotation, initially set the low-level sensor and capture motor amperages.

1. Lock out the control panel at the rotary disconnect lever.

2. Remove the hopper boot end cap. *(Reference page 38 of this manual for how to remove the end cap if needed)* The end cap is removed during this motor rotation check to prevent any damage to the auger or end cap in the event the motor is rotating in the wrong direction.



If there is a center core in your unit, it will slide out of the end of the boot once the cap is removed. So, it will also need to be removed to perform the motor rotation check.

3. Put the selector switch in manual mode.

4. Press the start button and run the motor just long enough to look down into the hopper and verify the auger is turning counter-clockwise. It should appear as though the auger is traveling up towards the drive. ***See next page for reference image.***

5. Once the rotation has been verified as correct, turn off the power at the disconnect, lock out the unit, and reinstall the center core and hopper boot end cap.

6. Once center core and end cap are back in place you can remove the lock out and turn power back on.

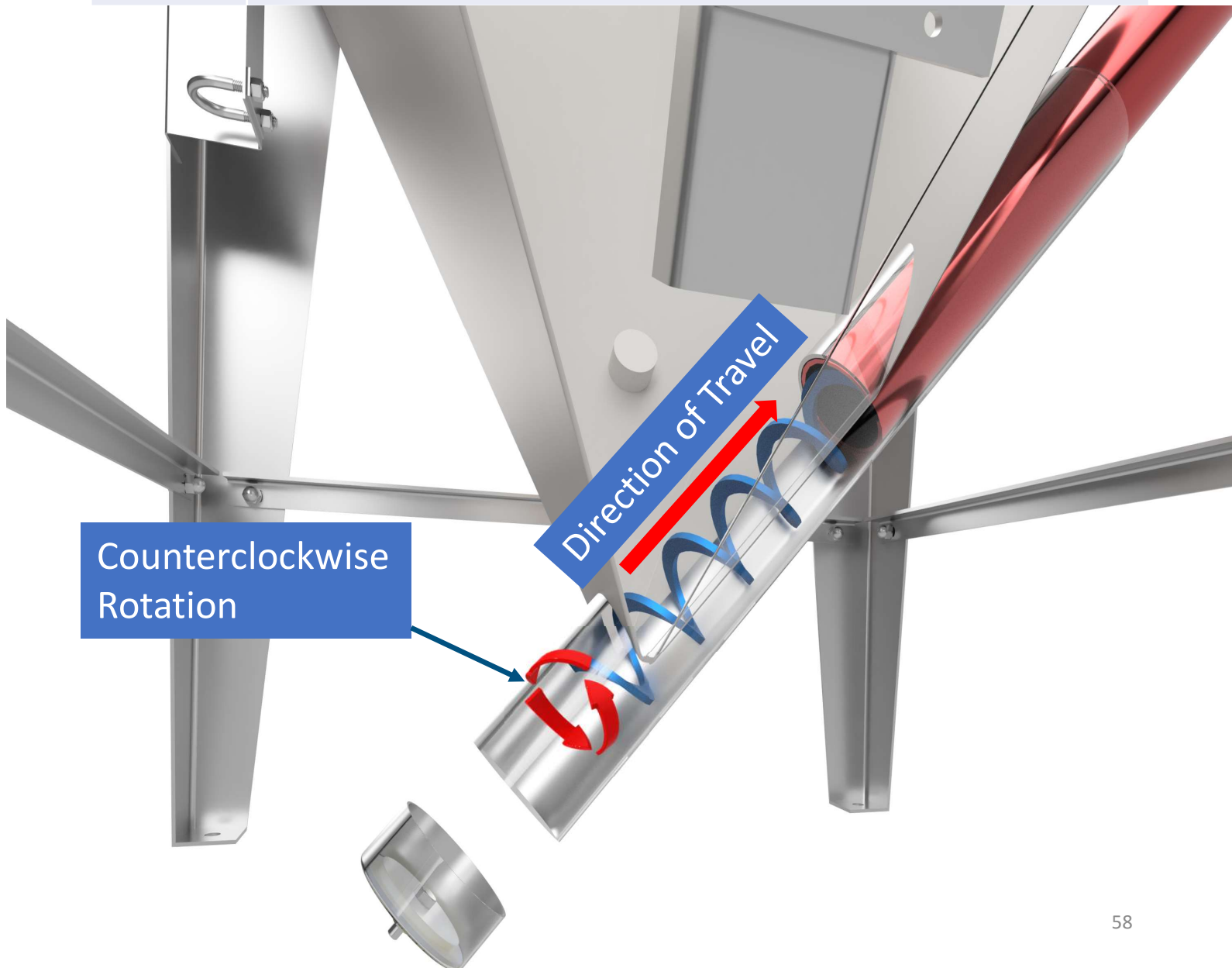
11.0 Start Up: Conveyor Checks with No Material (Motor Rotation)

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11.0 Start Up: Conveyor Checks with No Material (Low Level Sensor)

WARNING

Prevent serious injury or death.

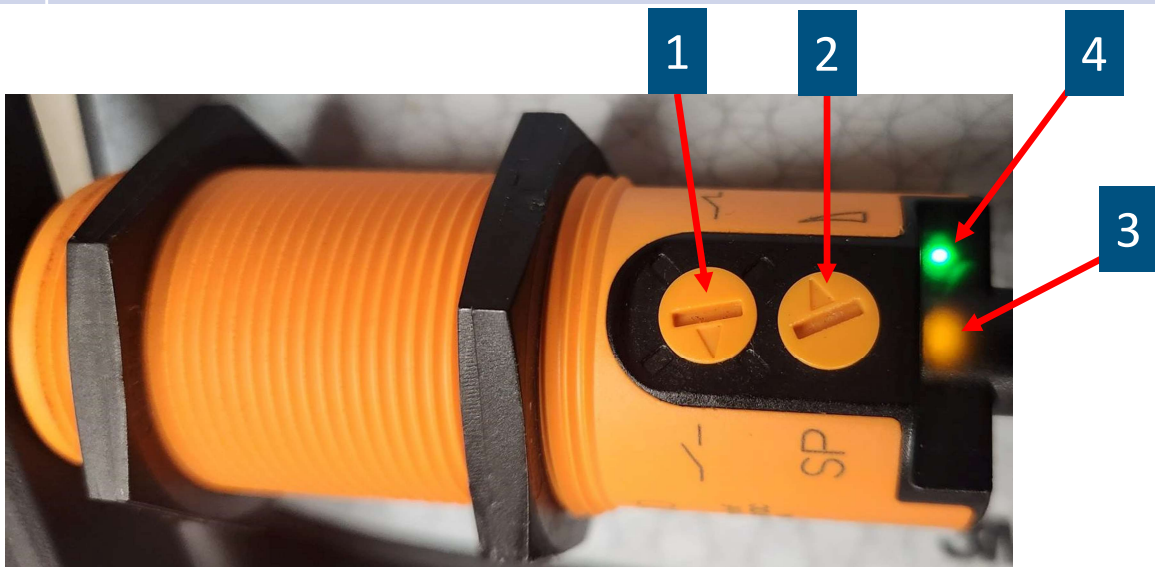
Electrical service installation must be performed by a qualified electrician and conform to all electrical codes.

The steps below are how to verify initial function of the standard low-level sensor provided by Hapman. The steps to calibrate the sensor will be called out in a later section.

NOTICE

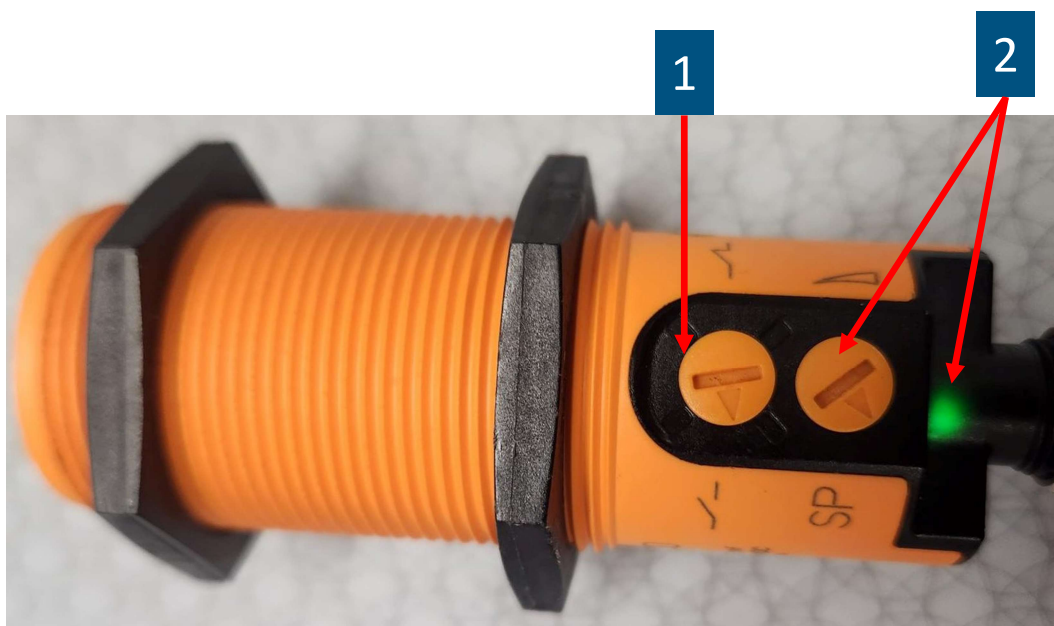
If your unit does not have a low-level sensor, then you can skip this step. If your low-level sensor is a paddle or probe style, please reference the vendor supplied instructions that came with your job specific Hapman manual.

1. Selector Switch to choose “Normally Open” (N.O.) or “Normally Closed” (N.C) function
2. Sensitivity Potentiometer, Clockwise for more sensitive, Counterclockwise for less sensitive
3. Yellow Light, indicates that the sensor is outputting a signal back to the control panel.
4. Green light indicates how close sensor is to the switch point, at which we activate the output (yellow light) on the sensor.



11.0 Start Up: Conveyor Checks with No Material (Low Level Sensor)

1. Verify the selector switch is pointing to the “Normally Open” (N.O.) position as indicated in the picture below
2. Using the sensitivity potentiometer adjust until only the green light is shown. This is the starting point for setting the sensor.
3. Now you will want to put your hand as close as you can to the face of the sensor and see if the green light moves to the right and the yellow light comes on. *(This may require removing the sensor from the hopper)*
4. When the yellow light is on verify that the correct input or relay is coming on in the panel.
5. Once this is verified put the sensor back in the hopper *(if removed for testing)*. Confirm once sensor is back in place that you are still seeing just the green light. If not, adjust sensitivity again until just the green light is showing.



11.0 Start Up: Conveyor Checks with No Material (Drive Motor Run Amperage)



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Severing and entanglement hazard.
Keep clear of conveyor and all moving parts.

Yaskawa VFD

1. Put the selector switch in manual mode.
2. Open the panel door so you can see the face of the VFD
3. Press the up arrow on the face of the VFD until you see 0.00A
4. Then press the start button on the front of the panel door
5. Let the unit run for 20 seconds and document the amperage being displayed on the front of the VFD. This is your base line amperage without material
6. Then press the stop button on the front of the panel.
7. Press the down arrow on the face of the VFD to return to the "F" screen.

NOTICE

If your unit has an Allen Bradley VFD please reference the next page for run amperage instructions.

11.0 Start Up: Conveyor Checks with No Material (Drive Motor Run Amperage)



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Severing and entanglement hazard.
Keep clear of conveyor and all moving parts.

Allen Bradley VFD

1. Put the selector switch in manual mode.
2. Open the panel door so you can see the face of the VFD
3. Press the ESC key, you see b001 come up on the display
4. Press the SEL button and move over to the last digit so that the 1 is flashing.
5. Press the up arrow until you see the number 3
6. Press enter, then press escape key
7. Then press the start button on the front of the panel door
8. Let the unit run for 20 seconds and document the amperage being displayed on the front of the VFD. This is your base line amperage without material.
9. Then press the stop button on the front of the panel door.
10. Repeat steps 3 & 4 and change b003 back to b001 to see preset speed.

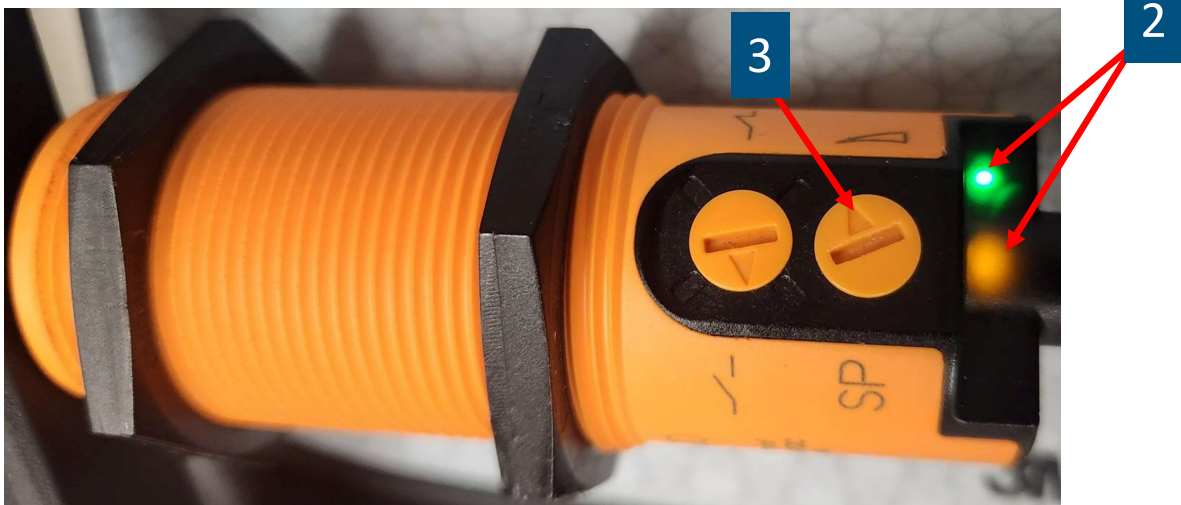
11.0 Start Up: Conveyor Checks with Material (Low Level Sensor)

Steps below are to perform the calibration of the low-level sensor with material present in the hopper.

1. Fill the hopper with material (*material level must be above the low-level sensor*)
2. Looking at the sensor you should see the yellow light on, and the green light moved out to the right.
3. If you do not see yellow light use the sensitivity potentiometer to increase the sensitivity until the yellow light comes on, and the green light moves to the right.
4. Put the Helix in “Auto Mode” on the selector switch located on the front of the control panel door and press start.
5. As the Helix runs and the material level in the hopper decreases the green light should move towards the yellow light. When the hopper is empty the yellow light should be off, and the green light should be on just like on the initial calibration. Once the level sensor yellow light goes out the Helix will stop running.

NOTICE

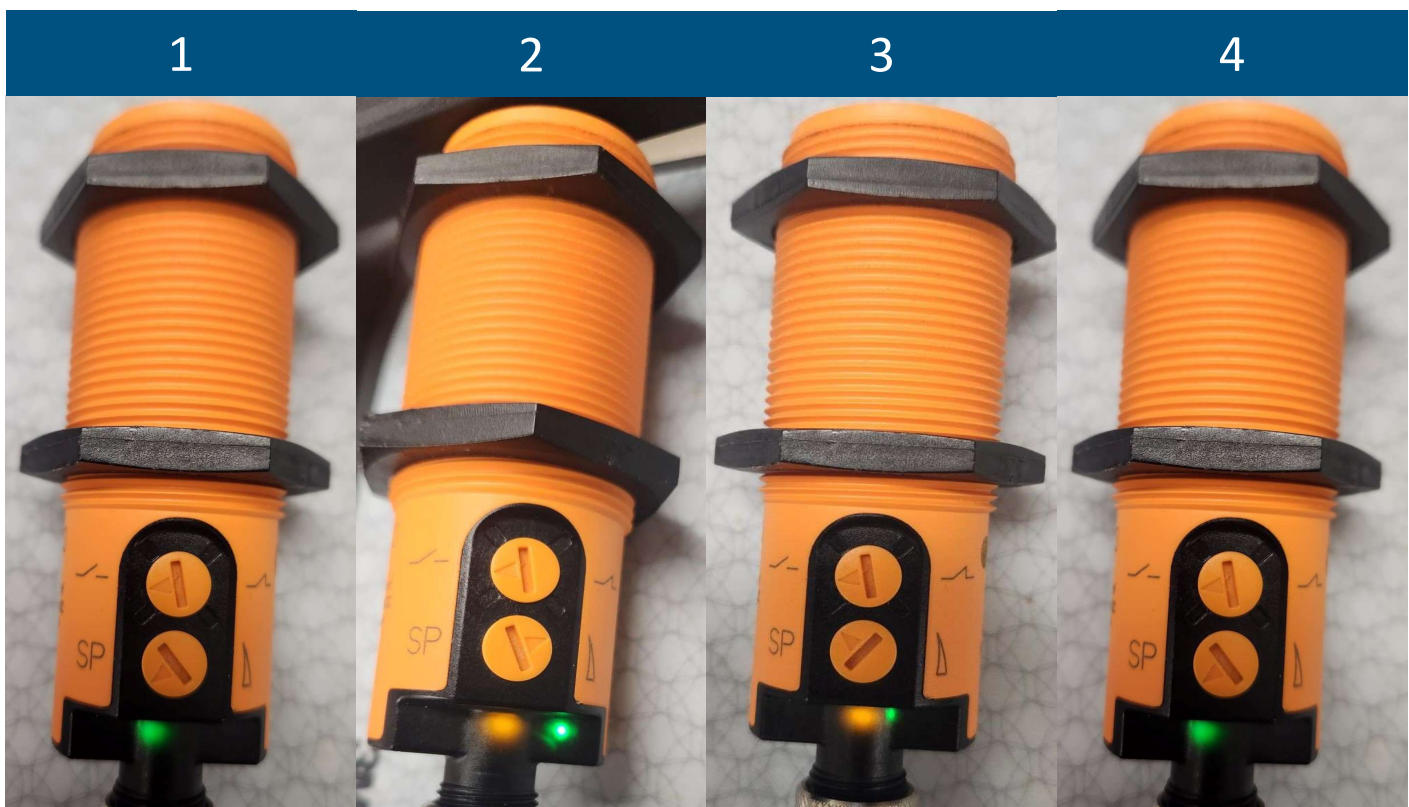
If multiple materials are being ran in the unit, then the low-level sensor may have to be readjusted if the material characteristics or bulk density are different.



11.0 Start Up: Conveyor Checks with Material (Low Level Sensor)

Below are visual examples of how the sensor should be reading during a regular running sequence.

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|----|---|
| 1. | Sensor when no material is present in the hopper, or the material level is below the sensor. The Helix will not start in “Auto Mode” when the sensor is in this state. |
| 2. | Sensor is seeing material (<i>yellow light is on</i>) and the green light has moved out to the right. |
| 3. | Helix is running and the material level is going down. The green light has moved closer to the yellow light indicating that it is about to switch off the output due to low material level. |
| 4. | The hopper material level is below the low-level sensor. The yellow light is off. Only the green light remains. The Helix will turn off at this point. |



11.0 Start Up: Conveyor Checks with Material (Drive Motor Run Amperage)

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⚠ WARNING

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Severing and entanglement hazard.
Keep clear of conveyor and all moving parts.

Yaskawa VFD

1. Put the selector switch in "Auto Mode".
2. Open the panel door so you can see the face of the VFD
3. Press the up arrow on the face of the VFD until you see 0.00A
4. Then press the start button on the front of the panel door
5. Let the unit run for 1-2 minutes and document the amperage being displayed on the front of the VFD. This is your base line amperage with material . Keep this documented value for future reference.
6. Then press the stop button on the front of the panel door.
7. Press the down arrow on the face of the VFD to return to the "F" screen.

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11.0 Start Up: Conveyor Checks with Material (Drive Motor Run Amperage)



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Allen Bradley VFD

1. Put the selector switch in "Auto" mode.
2. Open the panel door so you can see the face of the VFD
3. Press the ESC key, you see b001 come up on the display
4. Press the SEL button and move over to the last digit so that the 1 is flashing.
5. Press the up arrow until you see the number 3
6. Press enter, then press escape key
7. Then press the start button on the front of the panel door
8. Let the unit run for 1-2 minutes and document the amperage being displayed on the front of the VFD. This is your base line amperage with material. Keep this documented value for future reference.
9. Then press the stop button on the front of the panel door.
10. Repeat steps 3 & 4 and change b003 back to b001 to see preset speed.